STAT 464 Assignment 2 Question 1

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Load data

```
dow <- read.csv("dow.csv", header = TRUE)
head(dow)
```

```
##       Date    Close
## 1 01/04/10 10.58396
## 2 01/05/10 10.57202
## 3 01/06/10 10.57368
## 4 01/07/10 10.60686
## 5 01/08/10 10.61819
## 6 01/11/10 10.66399
```

a) Time Series Plot

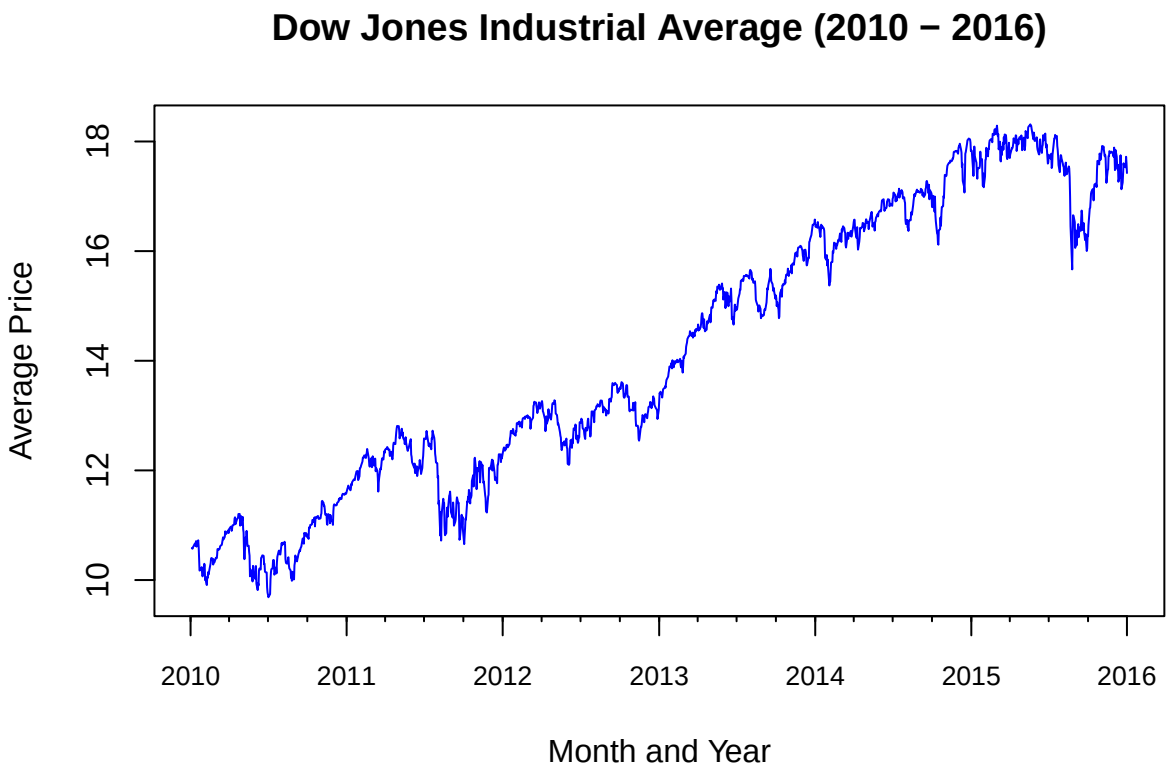
```
# Convert to R date in order to use seq properly
dow$Date <- as.Date(dow$Date, format = "%m/%d/%y")

quarterly_ticks <- seq(as.Date("2010-01-01"), max(dow$Date), by = "3 months")
jan_ticks <- seq(as.Date("2010-01-01"), as.Date("2015-01-01"), by = "1 year")
yearly_ticks <- c(jan_ticks, as.Date("2015-12-31"))
yearly_labels <- c("2010", "2011", "2012", "2013", "2014", "2015", "2016")

add_axis <- function() {
  axis(1, at = yearly_ticks, labels = yearly_labels, cex.axis = 0.8)
  axis(1, at = quarterly_ticks, labels = FALSE, tck = -0.01)
  # Added smaller tick marks for personal clarity
}

plot(dow$Date, dow$Close,
     type = "l",
     col = "blue",
     xaxt = "n",
     main = "Dow Jones Industrial Average (2010 - 2016)",
     xlab = "Month and Year",
     ylab = "Average Price")

add_axis()
```



Looks like a linearly-increasing trend overall. Just over $1 per year on average.

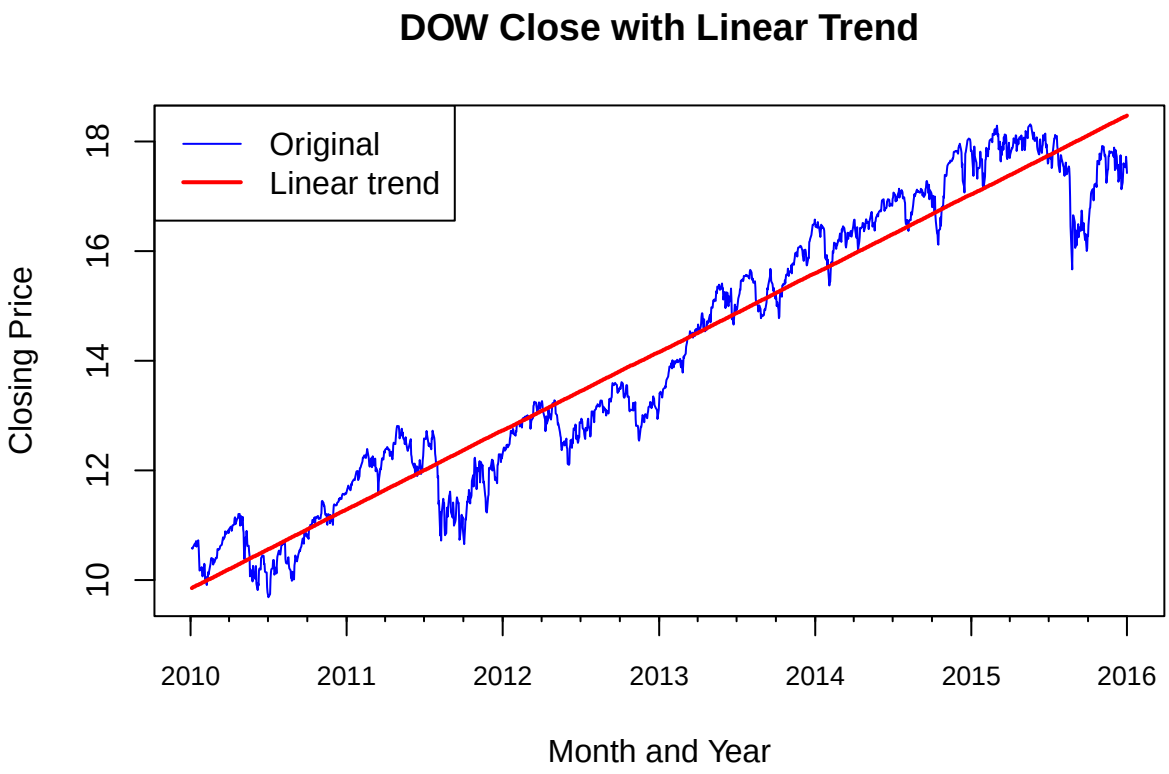
b) Linear Trend Fit

```
lin_trend <- trend(dow$Close, p = 1) # Linear Trend
lin_resid <- dow$Close - lin_trend
```

```
plot(dow$Date, dow$Close,
     type = "l",
     col = "blue",
     xaxt = "n",
     main = "DOW Close with Linear Trend",
     xlab = "Month and Year",
     ylab = "Closing Price")

lines(dow$Date, lin_trend, col = "red", lwd = 2)
add_axis()

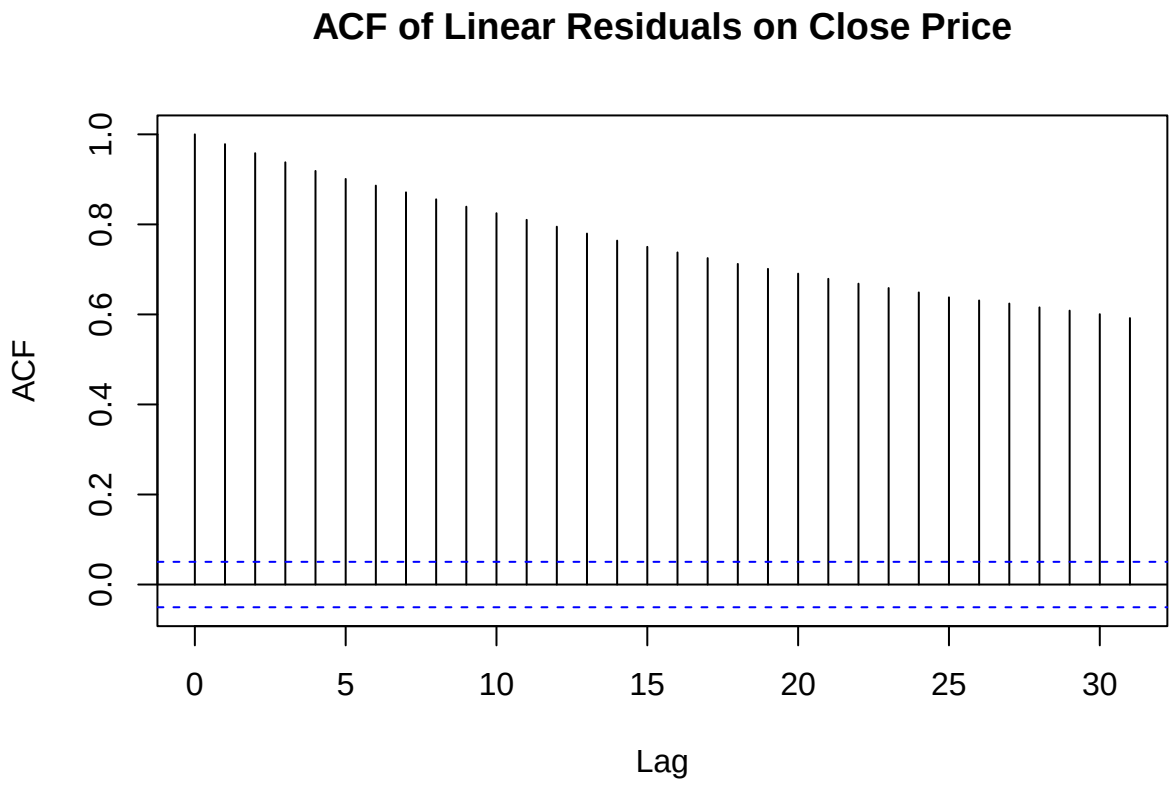
legend("topleft", legend = c("Original", "Linear trend"),
       col = c("blue", "red"), lty = 1, lwd = c(1, 2))
```



Given the randomness of the residuals and how jagged they are, they seem normally distributed like how a WN process would be. Hence, they appear like a WN process.

c) Sample ACF and Linear Regression Coefficient

```
lin_resid_acf <- acf(lin_resid,
                     main = "ACF of Linear Residuals on Close Price")
```




```r
r <- lin_resid # For consistency with the formula in the question
n <- length(r)
r_t  <- r[2:n] # Start at t = 2
r_t1 <- r[1:(n-1)]

model <- lm(r_t ~ r_t1 - 1) # - 1 will remove the intercept term

summary(model)
```

```
##
## Call:
## lm(formula = r_t ~ r_t1 - 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.65440 -0.06145  0.00481  0.06793  0.56756
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## r_t1 0.980065   0.005179   189.2   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1246 on 1508 degrees of freedom
## Multiple R-squared:  0.9596, Adjusted R-squared:  0.9596
## F-statistic: 3.582e+04 on 1 and 1508 DF,  p-value: < 2.2e-16
```

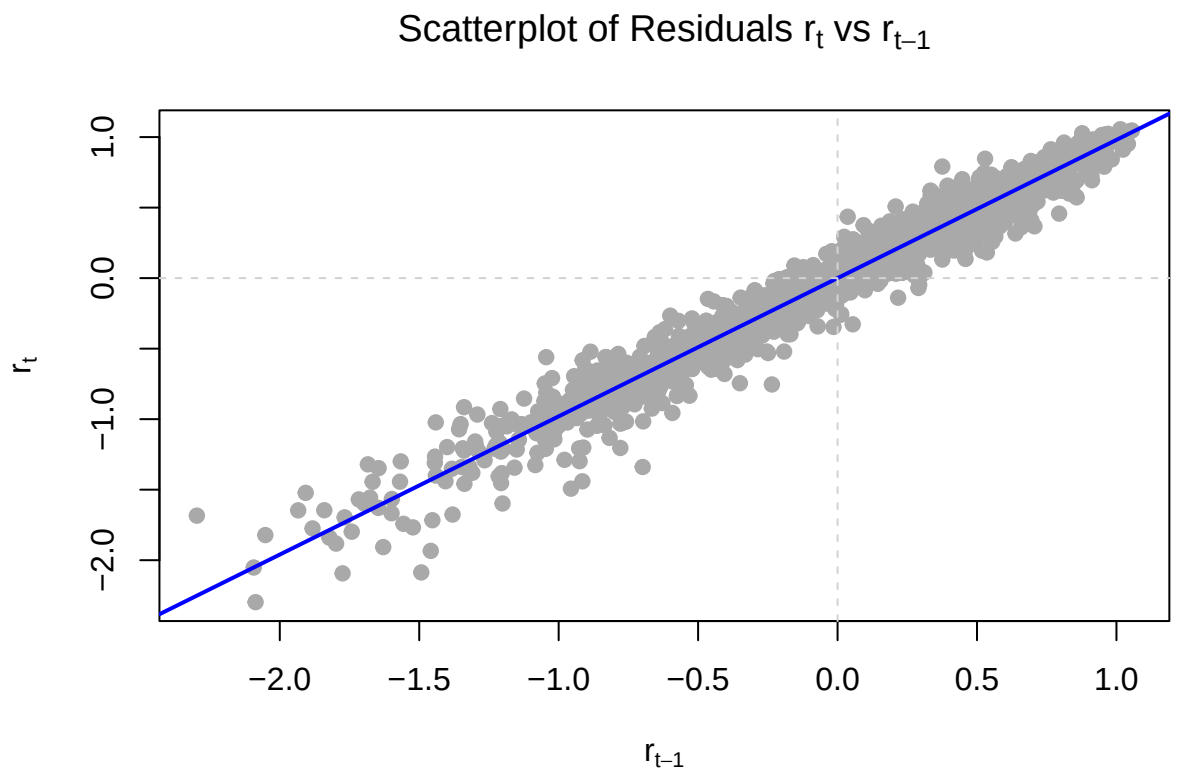
The estimated coefficient phi_hat is:

```
phi_hat <- coef(model)["r_t1"]
print(phi_hat)
```

```
##      r_t1
## 0.9800655
```

```
plot(r_t1, r_t,
     xlab = expression(r[t-1]),
     ylab = expression(r[t]),
     main = expression(paste("Scatterplot of Residuals ", r[t], " vs ", r[t-1])),
     pch = 19,
     col = "darkgray")

abline(model, col = "blue", lwd = 2)
abline(h = 0, v = 0, lty = 2, col = "lightgray")
```

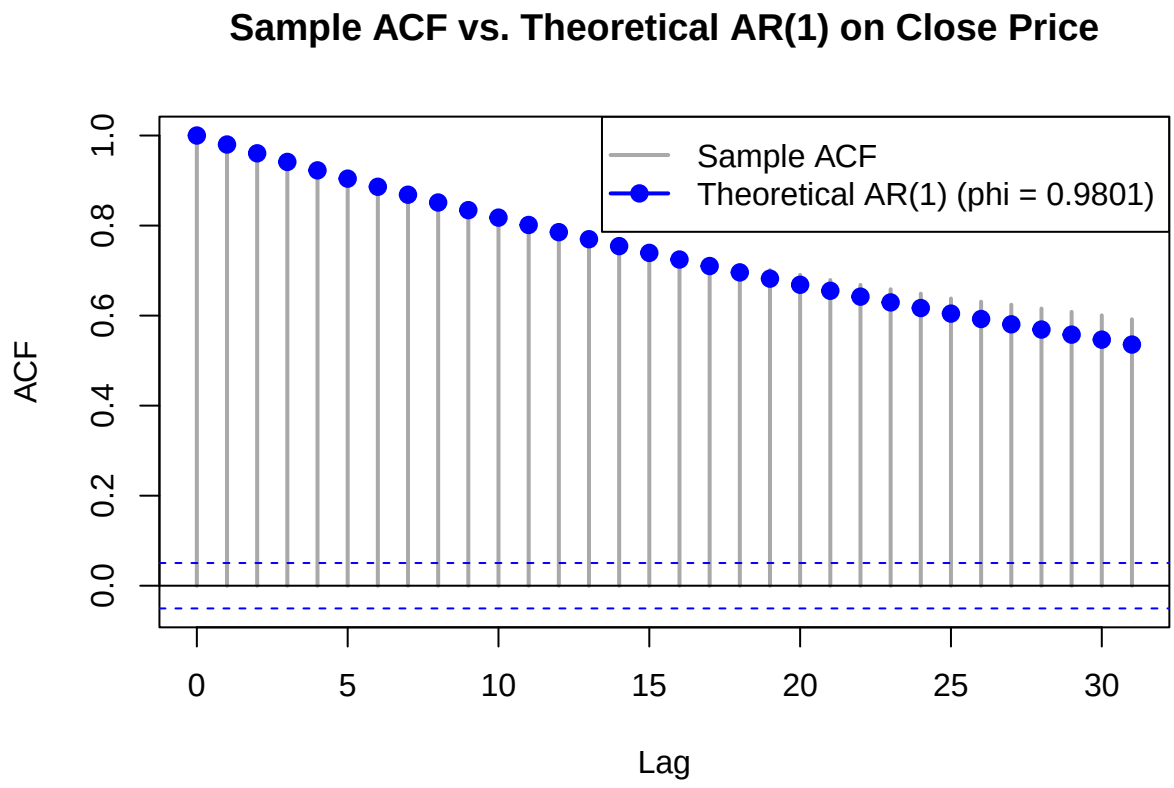



```
lin_resid_acf <- acf(lin_resid, na.action = na.pass,
                     main = "Sample ACF vs. Theoretical AR(1) on Close Price",
                     col = "darkgray", lwd = 2)

lags <- drop(lin_resid_acf$lag)
theoretical_acf <- phi_hat^lags
```

```r
lines(lags, theoretical_acf,
      type = "b",
      col = "blue",
      pch = 19,
      lwd = 2)

legend("topright",
       legend = c("Sample ACF",
                  paste0("Theoretical AR(1) (phi = ", round(phi_hat, 4), ")")),
       col = c("darkgray", "blue"),
       lty = 1,
       lwd = 2,
       pch = c(NA, 19))
```




```r
acf_orig <- acf(lin_resid, na.action = na.pass, plot = FALSE)
acf_ar1      <- acf(residuals(model), na.action = na.pass, plot = FALSE)

# Extract flat vectors
lags <- drop(acf_orig$lag)
res_orig <- drop(acf_orig$acf)
res_ar1 <- drop(acf_ar1$acf)

# Plot the original residuals
plot(lags, res_orig,
     type = "h",
```

```
     col = "darkgray",
     lwd = 1.5,
     # Adjust ylim to how big residuals are
     ylim = c(min(c(res_orig, res_ar1, -0.2)), 1),
     xlab = "Lag",
     ylab = "ACF",
     main = "ACF Comparison: Original vs. AR(1) Model Residuals")

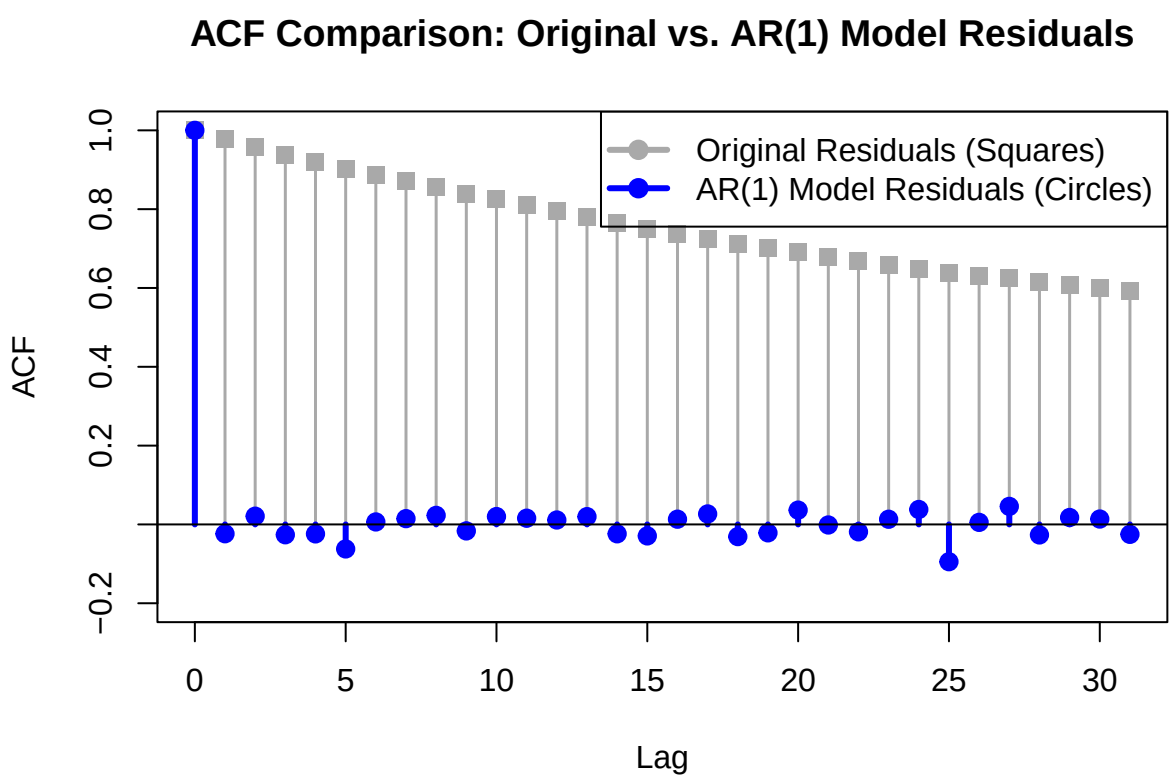
# Add squares at the top of the original ACF bars
points(lags, res_orig,
       pch = 15,
       col = "darkgray",
       cex = 1.2)

# Overlay the AR(1) model residual bars
lines(lags, res_ar1,
      type = "h",
      col = "blue",
      lwd = 3)

# Add circles at the top of the AR(1) ACF bars
points(lags, res_ar1,
       pch = 19,
       col = "blue",
       cex = 1.2)

# Add the x-axis
abline(h = 0, lty = 1, col = "black")

legend("topright",
       legend = c("Original Residuals (Squares)", "AR(1) Model Residuals (Circles)"),
       col = c("darkgray", "blue"),
       lty = 1,
       lwd = c(3, 3),
       pch = c(19, 19))
```



Seems like the AR(1) process' residuals show that the AR(1) process' method of accounting for the explained variance in the time series is strong, with a low correlation between residuals past the first lag. The remaining correlation would likely be stemming from any unexplained variance, or white noise, in the time series.

Compared to the original model's residuals, it is a much stronger fit. The original residuals seem to struggle to properly model the time series.